

GCSE Mathematics - Foundation Tier

Paper 2 (Calculator) Practice - Set 2

Time: 1 hour 30 minutes | Total marks: 80 | Calculator allowed

Instructions

- Use black ink or ball-point pen.
- Answer ALL questions in the spaces provided.
- You must show all your working - answers given without working may not score full marks.
- Diagrams are NOT accurately drawn unless otherwise stated.
- If your calculator does not have a pi button, use $\pi = 3.142$.

1.

Write these values in order of size. Start with the smallest.

$\frac{1}{3}$ 0.3 30% 0.33

(Total for Question 1 is 1 marks)

2.

Work out 35% of £320.

(Total for Question 2 is 2 marks)

3.

- (a) Convert 3.5 hours into minutes. (1)
- (b) Convert 270 minutes into hours and minutes. (1)

(Total for Question 3 is 2 marks)

4.

Write the ratio 45 : 60 in its simplest form.

(Total for Question 4 is 2 marks)

5.

A pictogram shows the number of cars sold at a small garage over five days. Each car symbol represents 4 cars.

Mon	[car][car][car]	(3 symbols)
Tue	[car][car][half-car]	(2.5 symbols)
Wed	[car][car][car][car][car]	(5 symbols)
Thu	[car][car]	(2 symbols)
Fri	[car][car][car][half-car]	(3.5 symbols)

- (a) How many cars were sold on Wednesday? (1)

(b) Work out the total number of cars sold over the five days.

(2)

(Total for Question 5 is 3 marks)

6.

Simplify $7x - 4 + 3x + 9$.

(Total for Question 6 is 2 marks)

7.

A solid wooden block has a volume of 250 cm^3 and is made of oak. The density of oak is 0.72 g/cm^3 .

Work out the mass of the block. Give your answer in grams.

(Total for Question 7 is 3 marks)

8.

Here are the shoe sizes of 9 students.

4 5 5 6 6 6 7 8 9

(a) Write down the median shoe size.

(1)

(b) Work out the range of the shoe sizes.

(1)

(c) Write down the mode.

(1)

(Total for Question 8 is 3 marks)

9.

A box of 84 sweets is shared between two children, Ella and Finn, in the ratio 3 : 4.

Work out the number of sweets that each child receives.

(Total for Question 9 is 3 marks)

10.

Solve $4(x + 3) = 28$.

You must show all your working.

(Total for Question 10 is 3 marks)

11.

The diagram shows an L-shape made from two rectangles. The L-shape has the following measurements: an outer rectangle 10 cm by 8 cm, with a rectangle of 6 cm by 5 cm removed from the top-right corner.

[Diagram: L-shape: outer rectangle 10 cm by 8 cm with a 6 cm by 5 cm rectangle removed from the top-right corner.]

Work out the total area of the L-shape. Give your answer in cm^2 .

12.

The pie chart shows how 120 people travel to work.

[Diagram: pie chart with sectors - Car: 180 deg, Bus: 90 deg, Train: 60 deg, Walk: 30 deg.]

- (a) How many people travel to work by bus? (2)
- (b) What percentage of people walk to work? (2)

(Total for Question 12 is 4 marks)

13.

The area A of a circle with radius r is given by $A = \pi \times r^2$.

- (a) Work out the area when $r = 6$ cm. Give your answer to 1 decimal place. Use $\pi = 3.142$. (2)

The cost C , in pounds, of hiring a van is given by $C = 25 + 0.4m$, where m is the number of miles driven.

- (b) Work out the cost when $m = 120$ miles. (2)

(Total for Question 13 is 4 marks)

14.

A coat originally costs £85. It is reduced by 18% in a sale.

- (a) Work out the sale price of the coat. (3)

Jay invests £2000 in an account paying 3% simple interest per year.

- (b) Work out the total amount of interest Jay receives after 4 years. (2)

(Total for Question 14 is 5 marks)

15.

Here are the first four terms of an arithmetic sequence.

5, 8, 11, 14, ...

- (a) Write down the next term of the sequence. (1)
- (b) Find an expression for the n th term of the sequence. (2)
- (c) Is 100 a term in this sequence? You must show your working. (2)

(Total for Question 15 is 5 marks)

16.

Sara and Tom share some money in the ratio 2 : 5. Tom receives £40 more than Sara.

Work out the total amount of money they share.

(Total for Question 16 is 4 marks)

17.

A teacher records the height (cm) and arm span (cm) of 9 students.

Height	150	154	158	160	163	166	170	173	178
Arm span	148	152	157	160	162	165	171	172	179

- On the grid provided, plot these data as a scatter graph. (2)
- Describe the correlation shown by the data. (1)
- Draw a line of best fit. (1)
- A different student has a height of 168 cm. Use your line of best fit to estimate the arm span. (1)

(Total for Question 17 is 5 marks)

18.

A solid cylinder has a radius of 5 cm and a height of 12 cm.

[Diagram: solid cylinder of radius 5 cm and height 12 cm.]

- Work out the volume of the cylinder. Give your answer to 1 decimal place. Use $\pi = 3.142$. (3)

The cylinder is made of brass with density 8.4 g/cm^3 .

- Work out the mass of the cylinder. Give your answer in kilograms to 2 decimal places. (2)

(Total for Question 18 is 5 marks)

19.

The table shows the number of books read by each of 30 students in a month.

Books	0	1	2	3	4	5
Freq.	4	7	9	6	3	1

- Draw an accurate bar chart to represent this information. Label both axes. (3)
- Work out the mean number of books read per student. Give your answer to 2 decimal places. (2)

(Total for Question 19 is 5 marks)

20.

The sum of three consecutive integers is 105.

Let the smallest integer be x .

- Show that $3x + 3 = 105$. (2)
- Solve the equation to find the value of x . (2)
- Write down the three integers. (1)

(Total for Question 20 is 5 marks)

21.

A supermarket sells two sizes of orange juice.

Small carton:	500 ml	for	£1.20
Large carton:	1.5 L	for	£3.30

- (a) Work out the cost per 100 ml of the small carton. (2)
- (b) Work out the cost per 100 ml of the large carton. (2)
- (c) Which carton is the better value for money? Give a reason for your answer. (2)

(Total for Question 21 is 6 marks)

22.

A class of 60 students chose to study either French or Spanish. The two-way table shows some of the data.

	Boys	Girls	Total
French	14	___	30
Spanish	___	17	___
Total	___	___	60

- (a) Complete the two-way table. (3)
- (b) A student is chosen at random from the class. Work out the probability that the student is a girl who chose Spanish. (2)

(Total for Question 22 is 5 marks)

TOTAL FOR PAPER: 80 MARKS

Mark Scheme

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Question 1

- Correct order: 0.3, 30%, $\frac{1}{3}$, 0.33 - B1 (because $0.3 = 30\%$, $\frac{1}{3} = 0.333\dots$, 0.33 lies between)

Total: 1 marks

Question 2

- 320×0.35 or $320 \times 35 / 100$ - M1
- £112 - A1

Total: 2 marks

Question 3

- (a) $3.5 \times 60 = 210$ minutes - B1
- (b) $270 / 60 = 4$ remainder 30 $\rightarrow$ 4 hours 30 minutes - B1

Total: 2 marks

Question 4

- Both parts divided by a common factor (e.g. $15 : 20$) - M1
- $3 : 4$ - A1

Total: 2 marks

Question 5

- (a) $5 \times 4 = 20$ - B1
- (b) Total symbols $3 + 2.5 + 5 + 2 + 3.5 = 16$; multiplied by 4 - M1
- (b) 64 (cars) - A1

Total: 3 marks

Question 6

- $10x$ or 5 collected correctly - M1
- $10x + 5$ - A1

Total: 2 marks

Question 7

- Mass = density $\times$ volume - M1
- 0.72×250 - M1
- 180 (g) - A1

Total: 3 marks

Question 8

- (a) Median = 6 - B1
- (b) Range = $9 - 4 = 5$ - B1
- (c) Mode = 6 - B1

Total: 3 marks

Question 9

- $84 / (3 + 4) = 12$ - M1
- Either 3×12 or 4×12 worked out - M1
- Ella 36 and Finn 48 - A1

Total: 3 marks

Question 10

- Either $x + 3 = 7$ (dividing by 4) or $4x + 12 = 28$ (expand) - M1
- Linear step to isolate x (e.g. $4x = 16$) - M1
- $x = 4$ - A1

Total: 3 marks

Question 11

- Whole rectangle $10 \times 8 = 80 \text{ cm}^2$ or split L into two rectangles - M1
- Removed rectangle $6 \times 5 = 30 \text{ cm}^2$ or second part area - M1
- $50 (\text{cm}^2)$ - A1

Total: 3 marks

Question 12

- (a) $90 / 360 \times 120$ - M1
- (a) 30 (people) - A1
- (b) $30 / 360 \times 100$ - M1
- (b) 8.33% (accept 8.3% to 1dp) - A1

Total: 4 marks

Question 13

- (a) $3.142 \times 6^2 (= 3.142 \times 36)$ - M1
- (a) 113.1 (cm^2) accept 113.0 to 113.2 - A1
- (b) $25 + 0.4 \times 120$ - M1
- (b) £73 - A1

Total: 4 marks

Question 14

- (a) 18% of 85 = 15.30 or multiplier 0.82 - M1
- (a) $85 - 15.30$ or 85×0.82 - M1
- (a) £69.70 - A1
- (b) $2000 \times 0.03 \times 4$ or 60×4 - M1
- (b) £240 - A1

Total: 5 marks

Question 15

- (a) 17 - B1
- (b) Common difference 3 noted ($3n + \dots$) - M1
- (b) $3n + 2$ - A1
- (c) $3n + 2 = 100 \rightarrow n = 98/3$ not an integer - M1
- (c) States 100 is NOT in the sequence with reason - A1

Total: 5 marks

Question 16

- Difference of parts = $5 - 2 = 3$ parts - M1
- One part = $40 / 3$ - M1
- Total parts = 7, so $7 \times (40/3)$ - M1
- Total = £93.33 (accept £93.34 or exact fraction $280/3$) - A1

Total: 4 marks

Question 17

- (a) All 9 points plotted correctly - B2 (B1 if 6 or more correct)
- (b) Positive (correlation, strong) - B1
- (c) Sensible line of best fit drawn - B1
- (d) Estimate from line at height 168 cm (accept 166 - 172) - B1

Total: 5 marks

Question 18

- (a) $\pi \times 5^2 \times 12$ written or attempted - M1
- (a) $3.142 \times 25 \times 12$ evaluated - M1
- (a) 942.6 (cm³) accept 942.5 to 942.7 - A1
- (b) $942.6 \times 8.4 (= 7917.84 \text{ g})$ and $/ 1000$ - M1
- (b) 7.92 (kg) - A1

Total: 5 marks

Question 19

- (a) Axes correctly labelled (Books read / Frequency) with appropriate scale - B1
- (a) Heights drawn for 6 bars - B2 (B1 for 4 or 5 correct)
- (b) $\text{Sum}(fx) = 0 + 7 + 18 + 18 + 12 + 5 = 60$ - M1
- (b) Mean = $60 / 30 = 2.00$ - A1

Total: 5 marks

Question 20

- (a) Three integers identified as $x, x+1, x+2$ - M1
- (a) $x + (x+1) + (x+2) = 3x + 3 = 105$ shown clearly - C1
- (b) $3x = 102$ - M1
- (b) $x = 34$ - A1
- (c) 34, 35, 36 - B1

Total: 5 marks

Question 21

- (a) $1.20 / 5$ (cost per 100 ml in small) - M1
- (a) £0.24 per 100 ml - A1
- (b) $1.5 \text{ L} = 1500 \text{ ml}$ and $3.30 / 15$ (cost per 100 ml in large) - M1
- (b) £0.22 per 100 ml - A1
- (c) Comparison stated and large carton chosen - M1
- (c) Large carton is better value, with reason (lower cost per 100 ml) - A1

Total: 6 marks

Question 22

- (a) French girls = $30 - 14 = 16$ - B1
- (a) Spanish total = $60 - 30 = 30$; Spanish boys = $30 - 17 = 13$ - M1
- (a) All totals consistent: boys total 27, girls total 33 - A1
- (b) $17 / 60$ seen - M1
- (b) $17/60$ - A1

Total: 5 marks

Worked Solutions

Foundation Tier - Paper 2 (Calculator) - Set 2 | Detailed explanations

Question 1

Express each value as a decimal so they can be compared.

$$1/3 = 0.333\dots$$

$$0.3 = 0.300$$

$$30\% = 30/100 = 0.300$$

$$0.33 = 0.330$$

So $0.3 = 30\%$ (equal), then 0.33 , then $1/3$.

Order: 0.3 , 30% , 0.33 , $1/3$. (0.3 and 30% are equal; you can list them either way provided both come before 0.33 .)

Question 2

35% as a decimal = 0.35 .

$$0.35 \times 320 = 112.$$

Answer: £112.

Mental check: 10% of $320 = £32$, so $30\% = £96$, and $5\% = £16$. Total £112.

Question 3

(a) 1 hour = 60 minutes, so 3.5 hours = $3.5 \times 60 = 210$ minutes.

(b) $270 / 60 = 4.5$, i.e. 4 whole hours and 0.5 of an hour.

0.5 hour = 30 minutes, so 4 hours 30 minutes.

Question 4

HCF of 45 and 60. Factors of 45: 1, 3, 5, 9, 15, 45. Factors of 60: 1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60. HCF = 15.

Divide both by 15: $45/15 = 3$, $60/15 = 4$.

Answer: 3 : 4.

Question 5

(a) 5 symbols on Wednesday and each symbol = 4 cars, so $5 \times 4 = 20$ cars.

(b) Add the symbols across the week: $3 + 2.5 + 5 + 2 + 3.5 = 16$ symbols.

Total cars = $16 \times 4 = 64$.

Question 6

Collect like terms.

$$x\text{-terms: } 7x + 3x = 10x.$$

$$\text{Constants: } -4 + 9 = +5.$$

Answer: $10x + 5$.

Question 7

Mass = density $\times$ volume.

Mass = $0.72 \times 250 = 180$.

Answer: 180 g.

Question 8

(a) Median is the middle value of the 9 sorted values. The 5th value is 6.

(b) Range = largest - smallest = $9 - 4 = 5$.

(c) Mode is the most common value. 6 appears three times, the most.

Question 9

Total parts = $3 + 4 = 7$. One part = $84 / 7 = 12$ sweets.

Ella: $3 \times 12 = 36$ sweets.

Finn: $4 \times 12 = 48$ sweets.

Check: $36 + 48 = 84$.

Question 10

Method 1 - divide by 4 first.

$4(x + 3) = 28 \rightarrow x + 3 = 7 \rightarrow x = 4$.

Method 2 - expand first.

$4x + 12 = 28 \rightarrow 4x = 16 \rightarrow x = 4$.

Check: $4(4 + 3) = 4 \times 7 = 28$.

Question 11

Split the L-shape into two rectangles or subtract the missing rectangle.

Method - subtract: outer rectangle $10 \times 8 = 80 \text{ cm}^2$; missing rectangle $6 \times 5 = 30 \text{ cm}^2$; L-shape = $80 - 30 = 50 \text{ cm}^2$.

Answer: 50 cm^2 .

Question 12

(a) Bus sector = 90 deg out of 360 deg.

People = $(90 / 360) \times 120 = 0.25 \times 120 = 30$.

(b) Walk sector = 30 deg.

Percentage = $(30 / 360) \times 100 = 8.33\ldots\%$, i.e. 8.3% to 1 dp.

Question 13

(a) $A = \pi \times r^2 = 3.142 \times 6^2 = 3.142 \times 36 = 113.112$.

Rounded to 1 dp: 113.1 cm^2 .

(b) $C = 25 + 0.4 \times 120 = 25 + 48 = 73$.

Answer: £73.

Question 14

(a) $18\% \text{ of } £85 = 0.18 \times 85 = £15.30$.

Sale price = $85 - 15.30 = £69.70$. (Or $0.82 \times 85 = £69.70$.)

(b) Simple interest = principal $\times$ rate $\times$ time = $2000 \times 0.03 \times 4 = £240$.

Question 15

(a) Common difference is $+3$. Next term = $14 + 3 = 17$.

(b) The n th term = (common difference) $\times n$ + (zeroth term).

Zeroth term = $5 - 3 = 2$. So n th term = $3n + 2$.

(c) If 100 were in the sequence then $3n + 2 = 100$, giving $n = 98/3 = 32.66\dots$

n is not a whole number, so 100 is NOT a term of the sequence.

Question 16

The difference between Tom and Sara is 5 parts - 2 parts = 3 parts.

3 parts = £40, so 1 part = $40 / 3 = £13.333\dots$

Total = 7 parts = $7 \times 40/3 = £280/3 = £93.33$ (to the nearest penny).

Question 17

(a) Plot 9 points (height on x-axis, arm span on y-axis).

(b) Points rise from lower-left to upper-right $\rightarrow$ positive correlation (and strong).

(c) Draw a straight line through the middle of the points.

(d) Read off the y-value at $x = 168 \text{ cm}$. Typical answer 166 - 172 cm.

Question 18

(a) Volume of cylinder = $\pi \times r^2 \times h = 3.142 \times 5^2 \times 12 = 3.142 \times 25 \times 12 = 942.6 \text{ cm}^3$.

(b) Mass = density $\times$ volume = $8.4 \times 942.6 = 7917.84 \text{ g}$.

Convert to kg: $7917.84 / 1000 = 7.91784 \text{ kg}$, i.e. 7.92 kg to 2 dp.

Question 19

(a) Draw axes (Number of books on x-axis, Frequency on y-axis) and 6 bars of heights 4, 7, 9, 6, 3, 1.

(b) Mean from frequency table: $\text{Sum}(fx) = 0 \times 4 + 1 \times 7 + 2 \times 9 + 3 \times 6 + 4 \times 3 + 5 \times 1 = 0 + 7 + 18 + 18 + 12 + 5 = 60$.

Total students $\text{Sum}(f) = 30$. Mean = $60 / 30 = 2.00$.

Question 20

(a) Three consecutive integers: x , $x+1$, $x+2$.

$$\text{Sum} = x + (x+1) + (x+2) = 3x + 3.$$

Given sum is 105, so $3x + 3 = 105$. (Shown.)

(b) Subtract 3: $3x = 102$. Divide by 3: $x = 34$.

(c) Integers are 34, 35, 36. Check: $34 + 35 + 36 = 105$.

Question 21

(a) Small carton: 500 ml is 5 lots of 100 ml. Cost per 100 ml = $1.20 / 5 = £0.24$.

(b) Large carton: 1.5 L = 1500 ml is 15 lots of 100 ml. Cost per 100 ml = $3.30 / 15 = £0.22$.

(c) $£0.22 < £0.24$, so the large carton is cheaper per 100 ml.

Conclusion: the large (1.5 L) carton is the better value.

Question 22

(a) French total is 30. French boys = 14, so French girls = $30 - 14 = 16$.

Spanish total = $60 - 30 = 30$. Spanish girls = 17, so Spanish boys = $30 - 17 = 13$.

Boys total = $14 + 13 = 27$. Girls total = $16 + 17 = 33$. Check: $27 + 33 = 60$.

(b) $P(\text{girl AND Spanish}) = (\text{Spanish girls}) / \text{total} = 17 / 60$.
